

periment) which will be sort of a white box test for the entire system. HEX1 will involve the RLV-TD taking off into space, making a re-entry and then finally landing at a designated spot.

ADITYA-L1

Packing six payloads of advanced measuring instruments, Aditya-L1 is the first of its kind from India to study the Sun's corona. The most important of these payloads is the Visible Emission Line Coronagraph (VELC). Aditya-L1 signified that it will be placed in earth's halo orbit at a position termed as L1 or Lagrangian 1. This position has the benefit of being completely unobstructed by any man-made object in space while still staying in orbit. It's about 1.5 million kilometers from earth.

Aditya-L1 with its bevy of instruments will observe the Sun's photosphere (soft and hard X-ray), corona (Visible and NIR) and Chromosphere (UV). There are also additional payloads to study the particle flux that is emanating from the Sun. Other payloads involve the Aditya Solar wind Particle Experiment (ASPEX) which will study solar wind. Then there's the Plasma Analyser Package for Aditya (PAPA) which will try to figure out the composition of solar wind and its energy distribution. As for studying X-ray flares, there's the Solar Low Energy X-ray Spectrometer (SoLEXS). In a similar vein, the High Energy L1 Orbiting X-ray Spectrometer (HELIOS) is designed to study the solar corona and estimate the energy

levels of various eruptive events. One of the benefits of being in L1 is that it is safe from earth's magnetic field and allows the onboard magnetometer to measure the magnitude and study the Interplanetary Magnetic Field.

Aditya-L1 is to be launched sometime around 2019-2020 by PSLV-XL from Sriharikota.

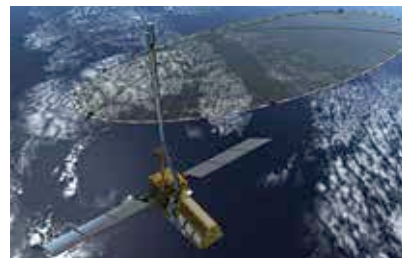
MANNED SPACE FLIGHT PROGRAMME

While a Manned Space Flight Programme does seem like the biggest project from among all that we've listed out on this page, the disappointing fact is that this is still a proposal. So without the Government green-lighting the proposal, ISRO will not be working towards realising the Indian Manned Space Flight Programme. It should be noted that most of the capabilities required to make this a possibility have already been realised. We have the GSLV-Mk III which has performed two launches successfully that is more than capable of carrying a two-man crew. However, it's very likely that a lot of modifications will be required to make it capable of taking humans to space. Speaking of which, our Astronauts will either be called 'Gaganauts' or 'Vyomanauts'.

The ISRO has already begun work into building a fully autonomous orbital vehicle to carry a two-man crew to Low Earth Orbit. A training facility for these astronauts will be set up in Bengaluru and other facilities to train and build capabilities towards making the Indian Manned Space Flight Programme a reality. All of this is based on quotes by different key personnel of ISRO at different conferences. There has been no proper announcement regarding the Manned Space Flight Programme so far. Current estimates point to the year 2024 as when we'll have all components for realising this programme in place. All that's needed is for the Government to green-light the proposal.

NISAR

A joint project between ISRO and NASA, NISAR stands for NASA-ISRO Synthetic Aperture Radar (NISAR) satellite. The aim of the project is to co-develop a dual frequency synthetic aperture radar satellite. Once launched, the satellite will be used to take measurements of the causes and consequences of land surface changes. In simple words, it's purpose is to study the effects of



(Credits: NASA/JPL Caltech)

Artist's rendition of NISAR

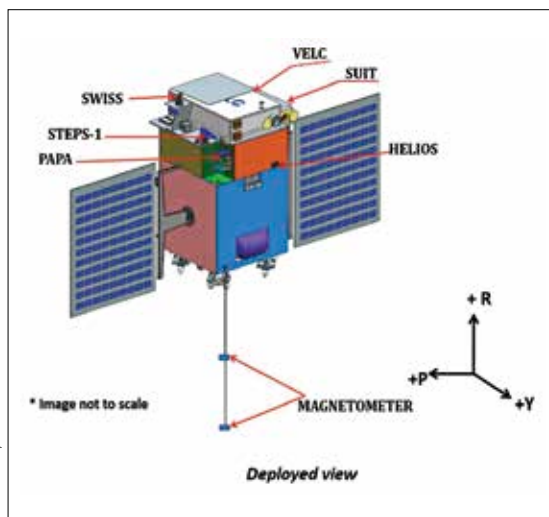
natural hazards such as earthquakes, tsunamis, volcanic eruptions, ice sheet collapse, etc.

The satellite will be the first to utilise two frequency bands, the first will be an L-band synthetic aperture radar (SAR) built by NASA and the other will be an S-band synthetic aperture radar built by ISRO. NASA will also provide a high-rate communication subsystem, GPS receivers, solid-state recorder, GPS receivers and a payload data subsystem, while ISRO will handle the satellite bus and the launch vehicle. The satellite will deploy a large (12-meter) mesh antenna which will operate on the two bands used by the individual SARs.

NISAR is expected to be launched sometime between 2020-21 from India into Low Earth Orbit with a planned life of 3.25 years.

THERE'S MORE TO COME

It should be pretty evident that ISRO has been a busy bee all these years. The RLV-TD and Manned Space Flight Programme has us stoked the most. And with more agencies and laboratories under the purview of the ISRO, we're destined to see many more such exciting projects to come out of ISRO's stables. Let us know which of these projects piqued your interest the most by writing to us at editor@digit.in. 



(Credits: ISRO)

The components on board Aditya-L1's main payload